

RestroPOS: POS Features Guide

This guide explains how each functional feature of the RestroPOS dashboard and client-side ordering flows operates.

1. POS & Quick Billing Terminal

The central interface for the cashier to take walk-in, takeaway, and dine-in orders quickly.

- **Cart Management:** Click-to-add items, live total calculation with tax rates (GST) and discount calculations.
- **KOT (Kitchen Order Ticket) Generation:** Instantly sends selected items to KOT status (PENDING -> PREPARING -> READY -> SERVED). Cashiers can print physical slips or view them digitally.
- **Split Settlement:** Cashiers can log payment methods (CASH, UPI, CARD, WALLET) and references for split payment bills before closing the order.

2. Dynamic Table Management

Allows management of the physical layout of the dining room.

- **Status Indicators:** Color-coded table states:
 - ■ **Green (Available):** Ready for guests.
 - ■ **Red (Occupied):** Actively ordering.
 - ■ **Blue (Reserved):** Booked.
 - ■ **Yellow (Cleaning):** Needs busing.
- **Interactive Floor Map:** Staff can change statuses, merge bills, or assign new orders to tables with a single tap.
- **Dynamic QR Code Generation:** Instantly generates a unique QR code for each table to be printed and pasted on table stands.

3. Zero-Trust QR Self-Ordering

The hardened, tamper-proof QR code ordering system designed to protect against URL abuse.

- **Ephemeral Session UUIDs:** Instead of embedding static database IDs in the QR code (which competitors do, enabling cross-table ordering hacks), the system creates a random, unique `TableSession` UUID.

- **120-Minute Expiry (TTL):** Sessions automatically expire after 2 hours or when checked out, preventing customers from ordering from home after leaving the restaurant.
- **Steward Activation PIN:** When customers scan the QR code, they must enter a **4-digit PIN** provided by the waiter to activate ordering. This prevents random online spoofing.

4. Menu & Catalog Control

A centralized dashboard to manage catalog items.

- **Vegetarian/Non-Vegetarian Indicators:** Visual dots (Green square for Veg, Red square for Non-Veg) for easy customer identification.
- **Real-Time Availability Toggles:** Out-of-stock items can be toggled off instantly, hiding them from the self-ordering QR menu.
- **Categories & Sort Orders:** Drag-and-drop hierarchy to prioritize high-margin dishes.

5. Inventory & Auto-Deduct Recipe Engine

Prevents stock discrepancy by linking menu dishes to raw materials.

- **Ingredient Links:** For example, selling a *Paneer Butter Masala* automatically deducts 200g of *Paneer* and 50g of *Butter* from the inventory.
- **Wastage Logs:** Tracks spoiled ingredients separate from sales.
- **Low-Stock Alerts:** Highlights items that fall below safety thresholds in red, preventing kitchen service issues.

6. Staff & Shift Monitoring

Manages profiles and logs for floor staff.

- **Shift Types:** Setup shift rotations (Morning, Evening, Full Day).
- **Check-In Log:** Tracks attendance status (PRESENT, ABSENT, LATE, ON_LEAVE) and automatically logs minutes late.
- **Leave Request Workflow:** Staff can submit requests; Managers review and approve them directly inside the dashboard.

7. Business Reports & GST Export

Deep analytics engine to monitor the restaurant's financial health.

- **GST Export Tool:** Generates formatted CSV spreadsheets containing transaction records, taxable amounts, CGST, SGST, and total bills for tax filing.
- **Source Breakdown:** Visual charts categorizing sales by source (DIRECT, SWIGGY, ZOMATO, TABLE_QR).
- **Revenue Trends:** Aggregates hourly/daily sales to highlight peak business times.

8. Live Simulator

A playground feature designed for staff training and testing.

- **Interactive QR Scan Mockup:** Simulates a guest scanning a table QR code in real time, entering the steward PIN, placing order items, watching KOT updates, and calling for the bill.
- **Zero Risk:** Allows testing of database isolation policies without touching live production metrics.